NYC yellow taxi trip data

DAMG7370 - Designing Advanced Data Architectures for Business Intelligence

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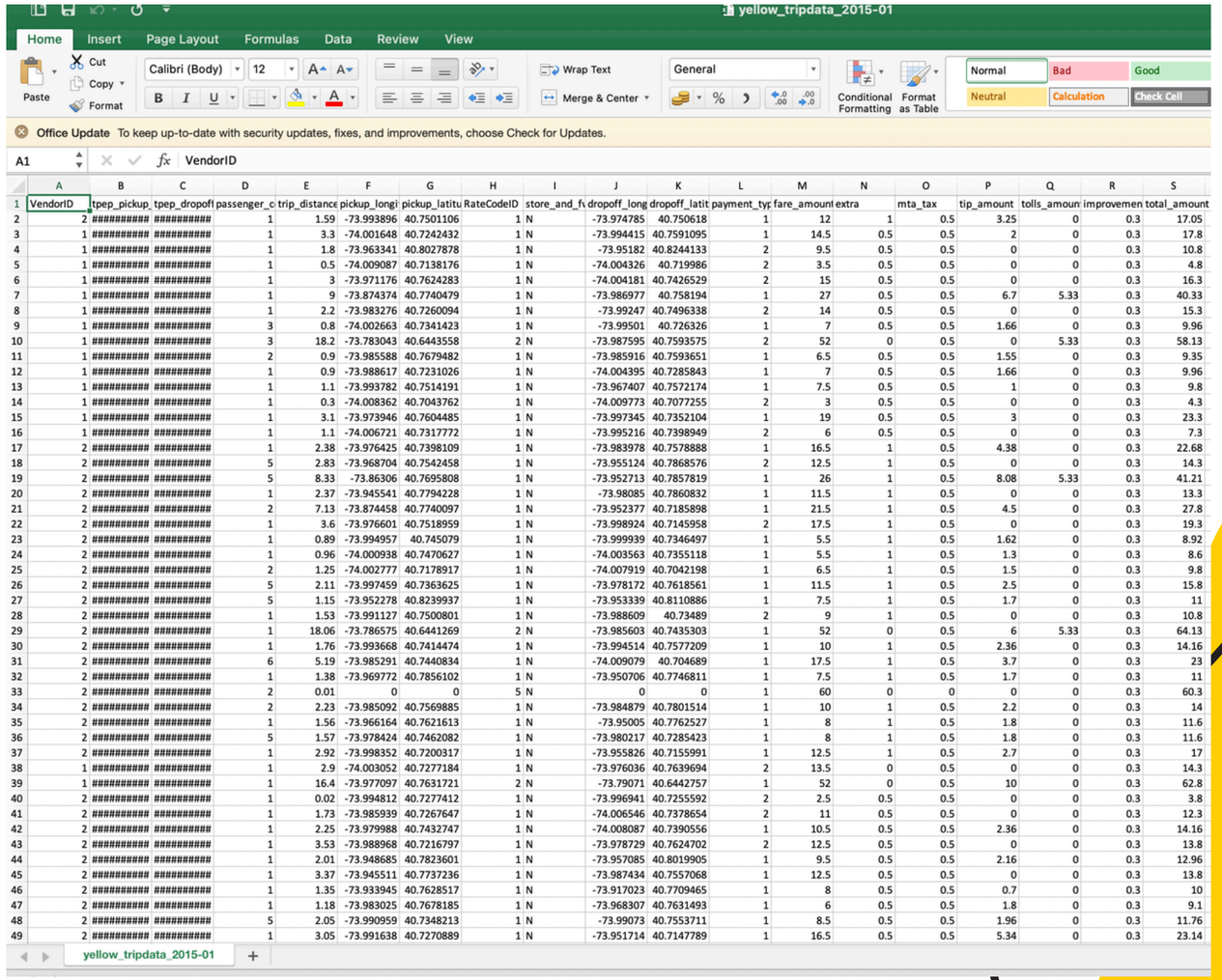
INTRODUCTION & DATASET OVERVIEW

Dataset Sources

- Source : NYC Taxi & Limousine Commission (TLC)
- Kaggle: NYC Yellow Taxi Trip Data
- Period: Jan 2015 & Jan-Mar 2016
- Size: 1.85 GB per CSV file
- Total rows: 12.7 million (cleaned)
- Columns: 19 per record

Key Columns

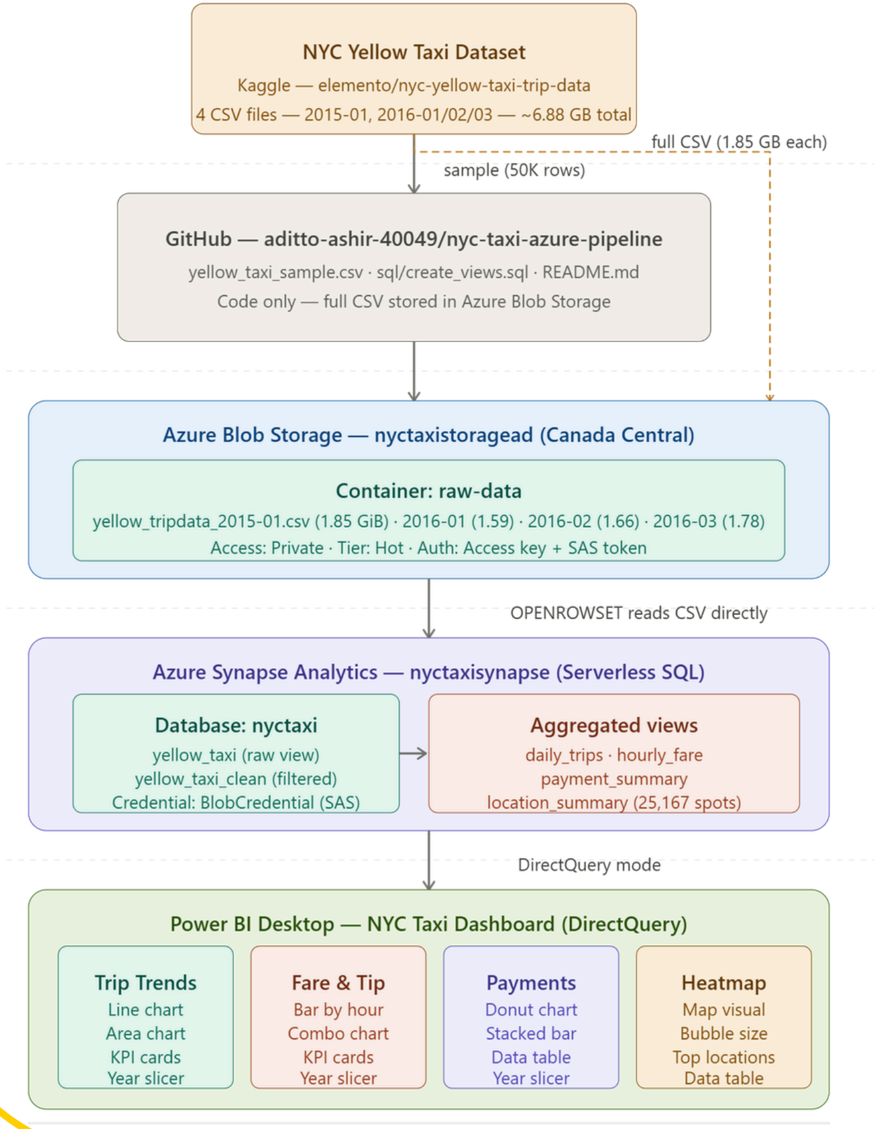
- Pickup & dropoff datetime
- Trip distance & duration
- Fare, tip & total amount
- Payment type



yellow_tripdata_2015-01

VendorID	tpep_pickup_	tpep_dropoff	passenger_c	trip_distance	pickup_longi	pickup_latitu	RateCodeID	store_and_fv	dropoff_long	dropoff_latit	payment_ty	fare_amount	extra	mta_tax	tip_amount	tolls_amoun	improvemen	total_amount
2	##########	##########	1	1.59	-73.993896	40.7501106	1	N	-73.974785	40.750618	1	12	1	0.5	3.25	0	0.3	17.05
1	##########	##########	1	3.3	-74.001648	40.7242432	1	N	-73.994415	40.7591095	1	14.5	0.5	0.5	2	0	0.3	17.8
1	##########	##########	1	1.8	-73.963341	40.8027878	1	N	-73.95182	40.8244133	2	9.5	0.5	0.5	0	0	0.3	10.8
1	##########	##########	1	0.5	-74.009087	40.7138176	1	N	-74.004326	40.719986	2	3.5	0.5	0.5	0	0	0.3	4.8
1	##########	##########	1	3	-73.971176	40.7624283	1	N	-74.004181	40.7426529	2	15	0.5	0.5	0	0	0.3	16.3
1	##########	##########	1	9	-73.874374	40.7740479	1	N	-73.986977	40.758194	1	27	0.5	0.5	6.7	5.33	0.3	40.33
1	##########	##########	1	2.2	-73.983276	40.7260094	1	N	-73.99247	40.7496338	2	14	0.5	0.5	0	0	0.3	15.3
1	##########	##########	3	0.8	-74.002663	40.7341423	1	N	-73.99501	40.726326	1	7	0.5	0.5	1.66	0	0.3	9.96
1	##########	##########	3	18.2	-73.783043	40.6443558	2	N	-73.987595	40.7593575	2	52	0	0.5	0	5.33	0.3	58.13
1	##########	##########	2	0.9	-73.985588	40.7679482	1	N	-73.985916	40.7593651	1	6.5	0.5	0.5	1.55	0	0.3	9.35
1	##########	##########	1	0.9	-73.988617	40.7231026	1	N	-74.004395	40.7285843	1	7	0.5	0.5	1.66	0	0.3	9.96
1	##########	##########	1	1.1	-73.993782	40.7514191	1	N	-73.967407	40.7572174	1	7.5	0.5	0.5	1	0	0.3	9.8
1	##########	##########	1	0.3	-74.008362	40.7043762	1	N	-74.009773	40.7077255	2	3	0.5	0.5	0	0	0.3	4.3
1	##########	##########	1	3.1	-73.973946	40.7604485	1	N	-73.997345	40.7352104	1	19	0.5	0.5	3	0	0.3	23.3
1	##########	##########	1	1.1	-74.006721	40.7317772	1	N	-73.995216	40.7398949	2	6	0.5	0.5	0	0	0.3	7.3
2	##########	##########	1	2.38	-73.976425	40.7398109	1	N	-73.983978	40.7578888	1	16.5	1	0.5	4.38	0	0.3	22.68
2	##########	##########	5	2.83	-73.968704	40.7542458	1	N	-73.955124	40.7868576	2	12.5	1	0.5	0	0	0.3	14.3
2	##########	##########	5	8.33	-73.86306	40.7695808	1	N	-73.952713	40.7857819	1	26	1	0.5	8.08	5.33	0.3	41.21
2	##########	##########	1	2.37	-73.945541	40.7794228	1	N	-73.98085	40.7860832	1	11.5	1	0.5	0	0	0.3	13.3
2	##########	##########	2	7.13	-73.874458	40.7740097	1	N	-73.952377	40.7185898	1	21.5	1	0.5	4.5	0	0.3	27.8
2	##########	##########	1	3.6	-73.976601	40.7518959	1	N	-73.998924	40.7145958	2	17.5	1	0.5	0	0	0.3	19.3
2	##########	##########	1	0.89	-73.994957	40.745079	1	N	-73.999939	40.7346497	1	5.5	1	0.5	1.62	0	0.3	8.92
2	##########	##########	1	0.96	-74.000938	40.7470627	1	N	-74.003563	40.7355118	1	5.5	1	0.5	1.3	0	0.3	8.6
2	##########	##########	2	1.25	-74.002777	40.7178917	1	N	-74.007919	40.7042198	1	6.5	1	0.5	1.5	0	0.3	9.8
2	##########	##########	5	2.11	-73.997459	40.7363625	1	N	-73.978172	40.7618561	1	11.5	1	0.5	2.5	0	0.3	15.8
2	##########	##########	5	1.15	-73.952278	40.8239937	1	N	-73.953339	40.8110886	1	7.5	1	0.5	1.7	0	0.3	11
2	##########	##########	1	1.53	-73.991127	40.7500801	1	N	-73.988609	40.73489	2	9	1	0.5	0	0	0.3	10.8
2	##########	##########	1	18.06	-73.786575	40.6441269	2	N	-73.985603	40.7435303	1	52	0	0.5	6	5.33	0.3	64.13
2	##########	##########	1	1.76	-73.993668	40.7414474	1	N	-73.994514	40.7577209	1	10	1	0.5	2.36	0	0.3	14.16
2	##########	##########	6	5.19	-73.985291	40.7440834	1	N	-74.009079	40.704689	1	17.5	1	0.5	3.7	0	0.3	23
2	##########	##########	1	1.38	-73.969772	40.7856102	1	N	-73.950706	40.7746811	1	7.5	1	0.5	1.7	0	0.3	11
2	##########	##########	2	0.01	0	0	5	N	0	0	1	60	0	0	0	0	0.3	60.3
2	##########	##########	2	2.23	-73.985092	40.7569885	1	N	-73.984879	40.7801514	1	10	1	0.5	2.2	0	0.3	14
2	##########	##########	1	1.56	-73.966164	40.7621613	1	N	-73.95005	40.7762527	1	8	1	0.5	1.8	0	0.3	11.6
2	##########	##########	5	1.57	-73.978424	40.7462082	1	N	-73.980217	40.7285423	1	8	1	0.5	1.8	0	0.3	11.6
2	##########	##########	1	2.92	-73.998352	40.7200317	1	N	-73.955826	40.7155991	1	12.5	1	0.5	2.7	0	0.3	17
1	##########	##########	1	2.9	-74.003052	40.7277184	1	N	-73.976036	40.7639694	2	13.5	0	0.5	0	0	0.3	14.3
1	##########	##########	1	16.4	-73.977097	40.7631721	2	N	-73.79071	40.6442757	1	52	0	0.5	10	0	0.3	62.8
2	##########	##########	1	0.02	-73.994812	40.7277412	1	N	-73.996941	40.7255592	2	2.5	0.5	0.5	0	0	0.3	3.8
2	##########	##########	1	1.73	-73.985939	40.7267647	1	N	-74.006546	40.7378654	2	11	0.5	0.5	0	0	0.3	12.3
2	##########	##########	1	2.25	-73.979988	40.7432747	1	N	-74.008087	40.7390556	1	10.5	0.5	0.5	2.36	0	0.3	14.16
2	##########	##########	1	3.53	-73.988968	40.7216797	1	N	-73.978729	40.7624702	2	12.5	0.5	0.5	0	0	0.3	13.8
2	##########	##########	1	2.01	-73.948685	40.7823601	1	N	-73.957085	40.8019905	1	9.5	0.5	0.5	2.16	0	0.3	12.96
2	##########	##########	1	3.37	-73.945511	40.7737236	1	N	-73.987434	40.7557068	1	12.5	0.5	0.5	0	0	0.3	13.8
2	##########	##########	1	1.35	-73.933945	40.7628517	1	N	-73.917023	40.7709465	1	8	0.5	0.5	0.7	0	0.3	10
2	##########	##########	1	1.18	-73.983025	40.7678185	1	N	-73.968307	40.7631493	1	6	0.5	0.5	1.8	0	0.3	9.1
2	##########	##########	5	2.05	-73.990959	40.7348213	1	N	-73.99073	40.7553711	1	8.5	0.5	0.5	1.96	0	0.3	11.76
2	##########	##########	1	3.05	-73.991638	40.7270889	1	N	-73.951714	40.7147789	1	16.5	0.5	0.5	5.34	0	0.3	23.14

ARCHITECTURE OVERVIEW



- **Source:** NYC TLC Yellow Taxi dataset from Kaggle — 4 CSV files (~6.88 GB total)
- **Version Control:** GitHub repo stores code, SQL scripts & 50K row sample (full CSVs too large for Git)
- **Storage:** Azure Blob Storage (nyctaxistoragead, Canada Central) — raw-data container holds all 4 CSV files
- **Processing:** Azure Synapse Analytics (Serverless SQL) reads CSVs directly via OPENROWSET — no data movement needed
- **Views:** 6 SQL views created in nyctaxi database — raw + cleaned + 4 aggregated views
- **Visualization:** Power BI Desktop connects via DirectQuery — 4 dashboard pages built on aggregated views

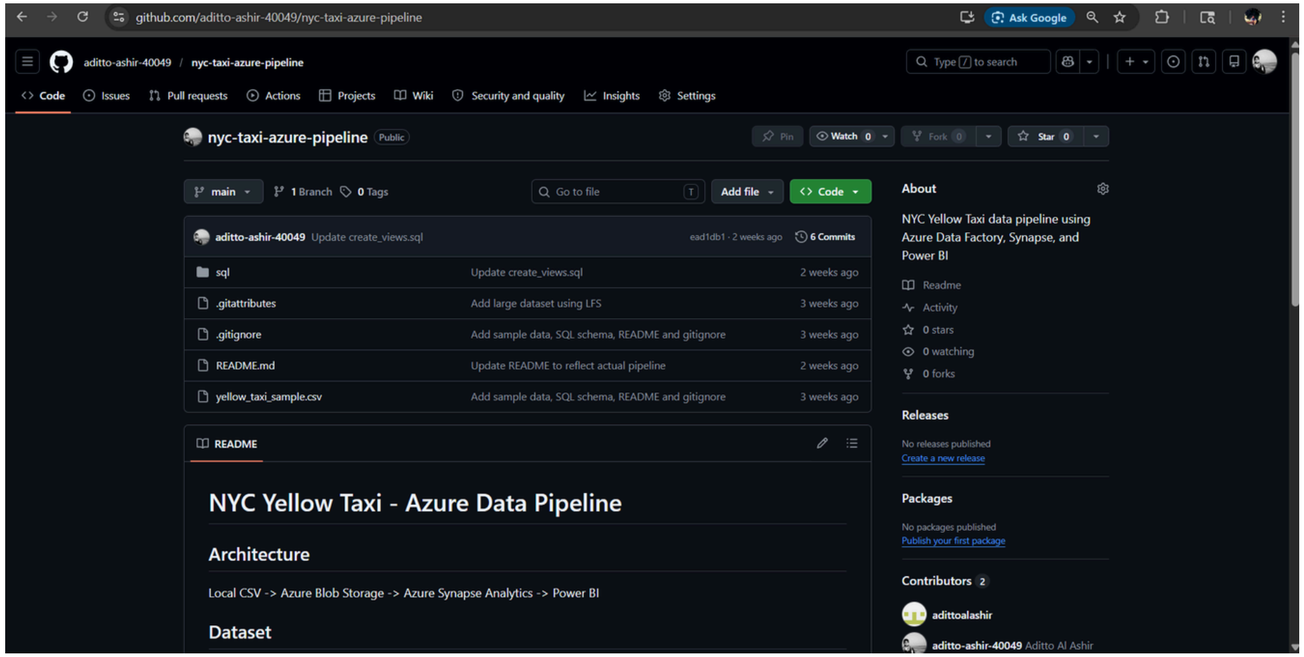
GITHUB SETUP

Repository Structure

- sql/ — All Synapse SQL scripts
- data/ — Sample dataset (50K rows)
- yellow_taxi_sample.csv — 50,000 row sample
- README with full pipeline documentation

Files Uploaded (Git LFS)

- yellow_tripdata_2015-01.csv
- yellow_tripdata_2016-01.csv
- yellow_tripdata_2016-02.csv
- yellow_tripdata_2016-03.csv

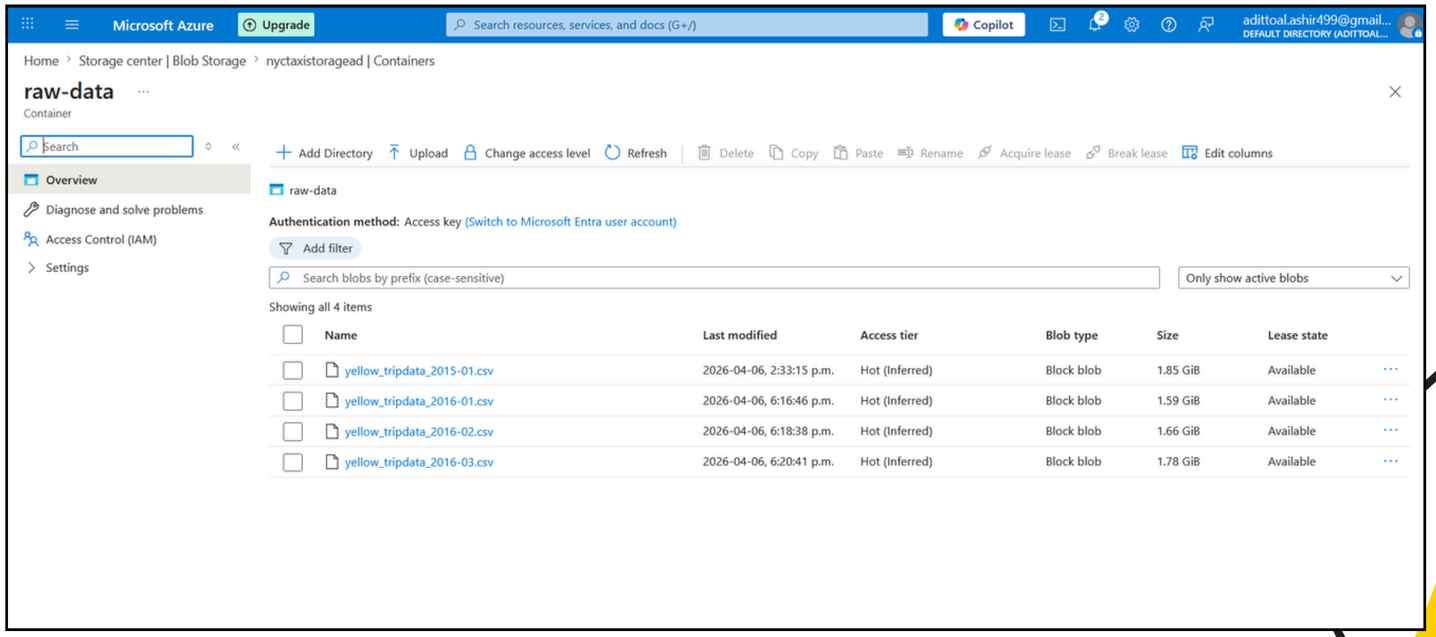


AZUR BLOB STORAGE

File	Size	Period
yellow_tripdata_2015-01.csv	1.85 GiB	Jan 2015
yellow_tripdata_2016-01.csv	1.59 GiB	Jan 2016
yellow_tripdata_2016-02.csv	1.66 GiB	Feb 2016
yellow_tripdata_2016-03.csv	1.78 GiB	Mar 2016

Storage Configuration

- Storage Account: nyctaxistoragead
- Region: Canada Central
- Container: raw-data
- Access Tier: Hot (Inferred)
- Auth: Access key + SAS token
- Total size: ~6.88 GB across 4 files

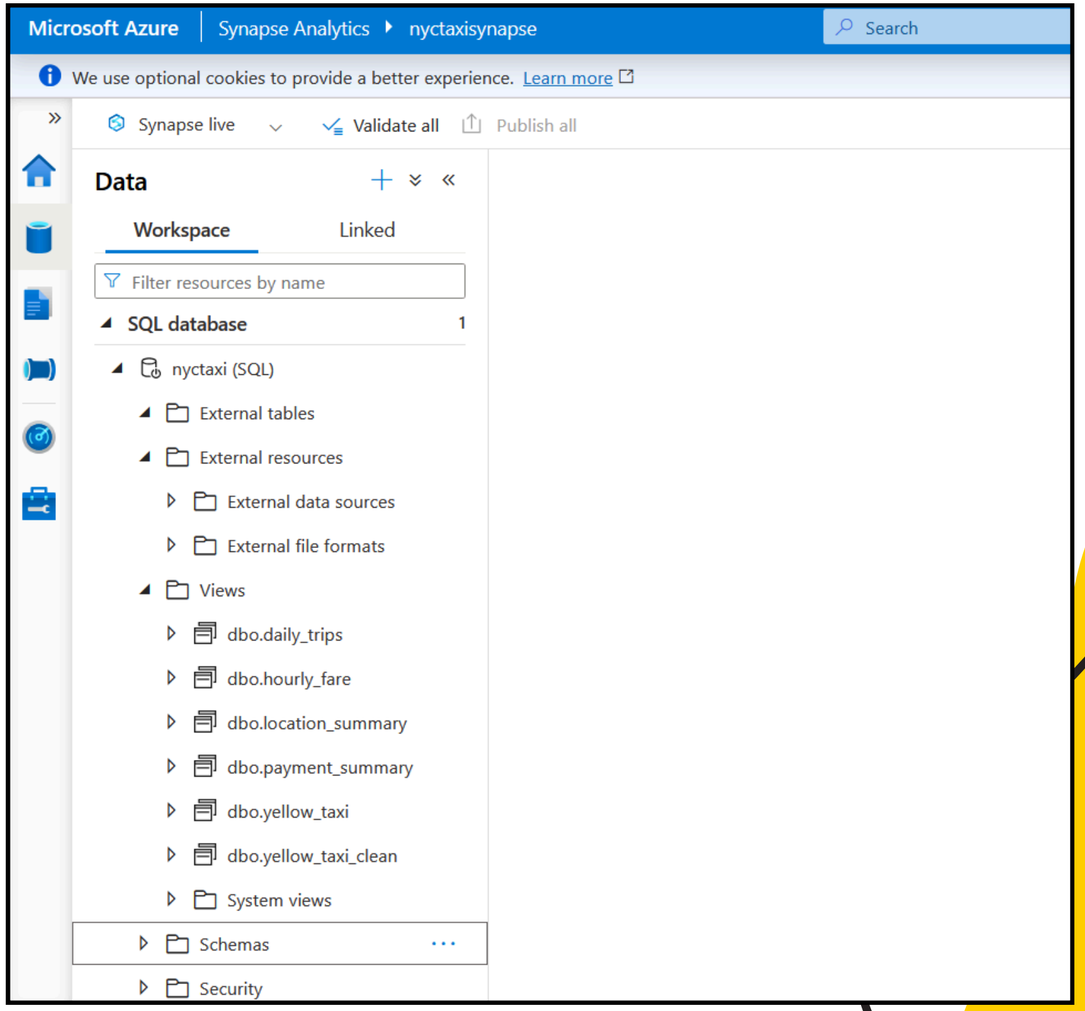


AZURE SYNAPSE ANALYTICS

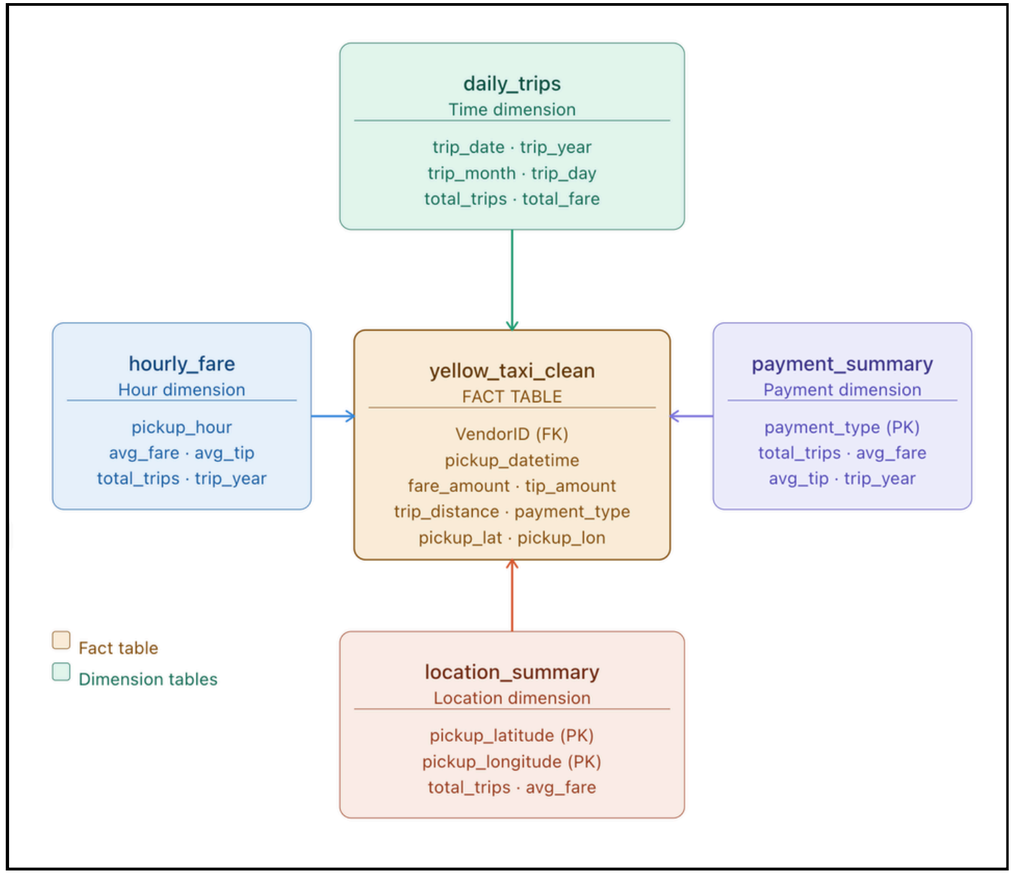
Views created:

- yellow_taxi — raw view combining all 4 CSV files via UNION ALL
- yellow_taxi_clean — filtered & validated data (12.7M rows)
- daily_trips — trip counts aggregated by date
- hourly_fare — fare & tip averages by hour of day
- payment_summary — breakdown by payment type
- location_summary — pickup hotspots (25,167 locations)

```sql
CREATE VIEW dbo.yellow_taxi AS
SELECT *, '2015-01' AS data_period
FROM OPENROWSET(
    BULK 'yellow_tripdata_2015-01.csv',
    DATA_SOURCE = 'BlobStorage',
    FORMAT = 'CSV',
    PARSER_VERSION = '2.0',
    HEADER_ROW = TRUE
) AS taxi
```



Star Schema Design



Fact Table
yellow_taxi_clean stores all 12.7M cleaned trip records with foreign keys linking to all 4 dimension views

daily_trips — Time Dimension
Aggregates trip counts and revenue by date — enables trend analysis across 2015 and 2016

hourly_fare — Hour Dimension
Groups fare and tip averages by hour of day — powers the peak hour and time analysis dashboards

payment_summary — Payment Dimension
Breaks down trips by payment type (credit card vs cash) — shows payment behavior trends by year

location_summary — Location Dimension
25,167 unique pickup coordinates — feeds the NYC heatmap showing Manhattan as the dominant zone

DATA CLEANING AND QUALITY

Issues Found	Rows Removed
Negative / zero fares	4,064
Zero distance trips	79,365
Zero passengers	6,565
Extreme fares (>$500)	77
Impossible distances (>100mi)	94

```sql
WHERE
    TRY_CAST(fare_amount AS FLOAT) > 0
AND TRY_CAST(fare_amount AS FLOAT) < 500
AND TRY_CAST(tip_amount AS FLOAT) >= 0
AND TRY_CAST(total_amount AS FLOAT) > 0
AND TRY_CAST(trip_distance AS FLOAT) > 0
AND TRY_CAST(trip_distance AS FLOAT) < 100
AND TRY_CAST(passenger_count AS INT) > 0
AND TRY_CAST(passenger_count AS INT) <= 6
AND TRY_CAST(payment_type AS INT) IN (1,2,3,4,5,6)
AND YEAR(CAST(tpep_pickup_datetime AS DATETIME2)) = 2015
AND CAST(tpep_dropoff_datetime AS DATETIME2) > CAST(tpep_pickup_datetime AS DATETIME2)
AND DATEDIFF(MINUTE,
```

12.7M
Rows Processed

90K+
Rows Removed

Cleaning Rules Applied

- Fare: $0.01 – $500
- Distance: 0.01 – 100 miles
- Passengers: 1 – 6
- Duration: 1 – 1440 mins
- Valid payment types: 1–6 only
- Dropoff must be after pickup
- Year must be 2015 or 2016

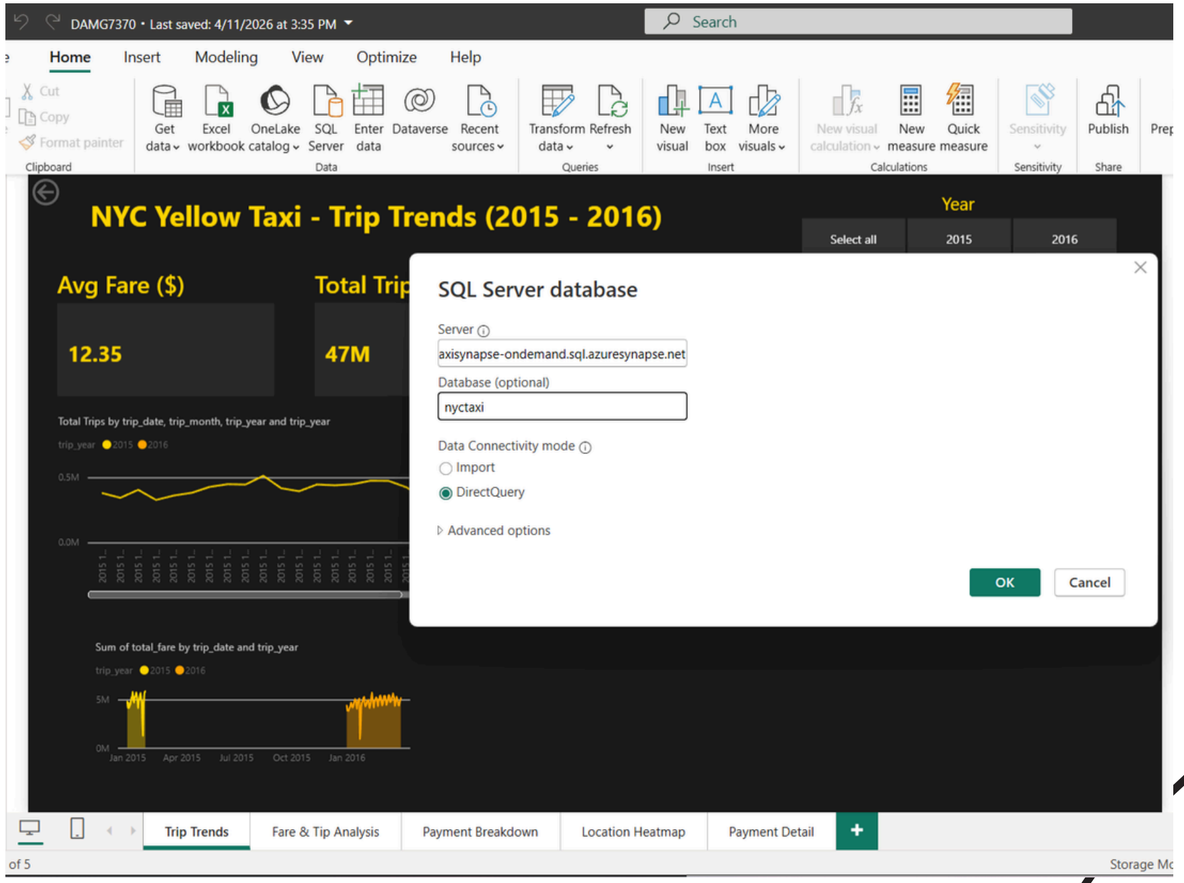
POWER BI CONNECTION

Connection Details

- Mode: DirectQuery
- Server: nyctaxisynapse-ondemand.sql.azuresynapse.net
- Database: nyctaxi
- Auth: Microsoft Account

5 Views Loaded into Power BI

- daily_trips
- hourly_fare
- payment_summary
- location_summary
- yellow_taxi_clean



DASHBOARD – TRIP BEHAVIOR & TIME ANALYSIS

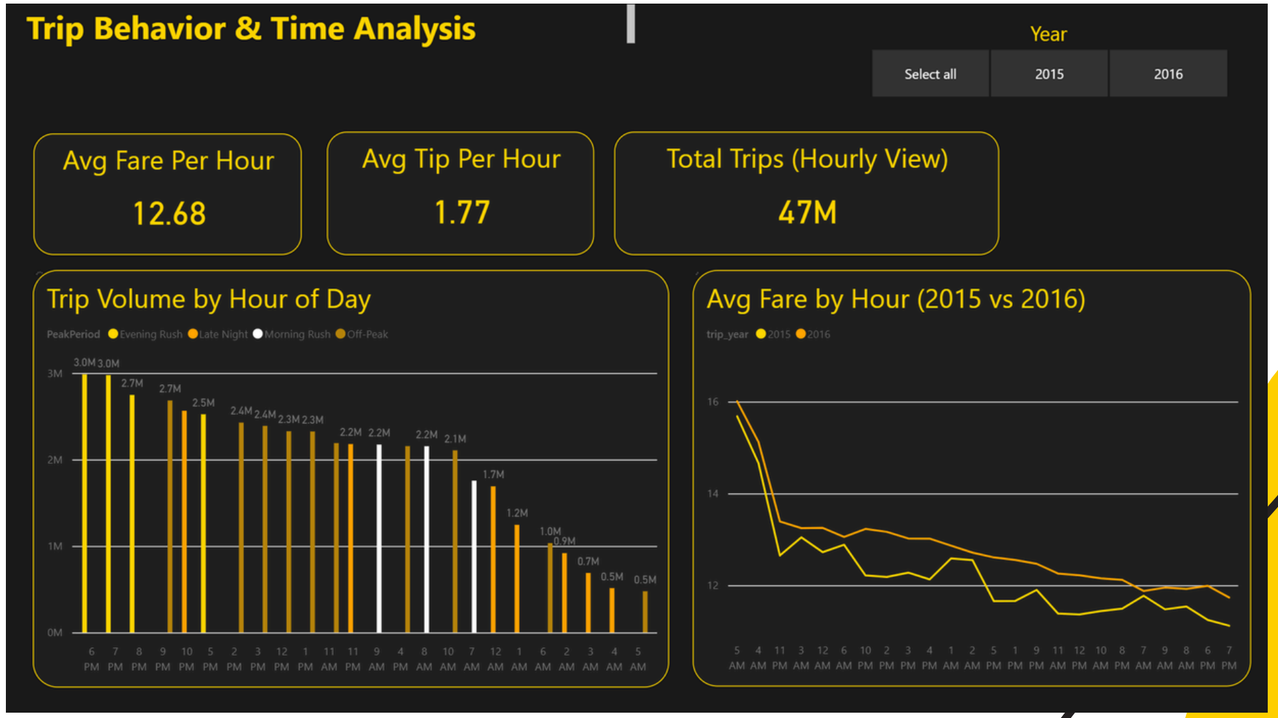
$12.68
Avg Fare Per Hour

$1.77
Avg Tip Per Hour

47M
Total Trips

Key Observations

- Evening rush (6–8 PM) peaks at 3M+ trips/hour
- Late night (3–5 AM) has the lowest volume
- Morning rush (8–10 AM) shows secondary peak
- 2015 fares start higher (~$16) and decline throughout the day
- 2016 shows a similar pattern at slightly lower fares
- Off-peak hours show the most stable fares (~$12)



DASHBOARD – FARE & TIP ANALYSIS

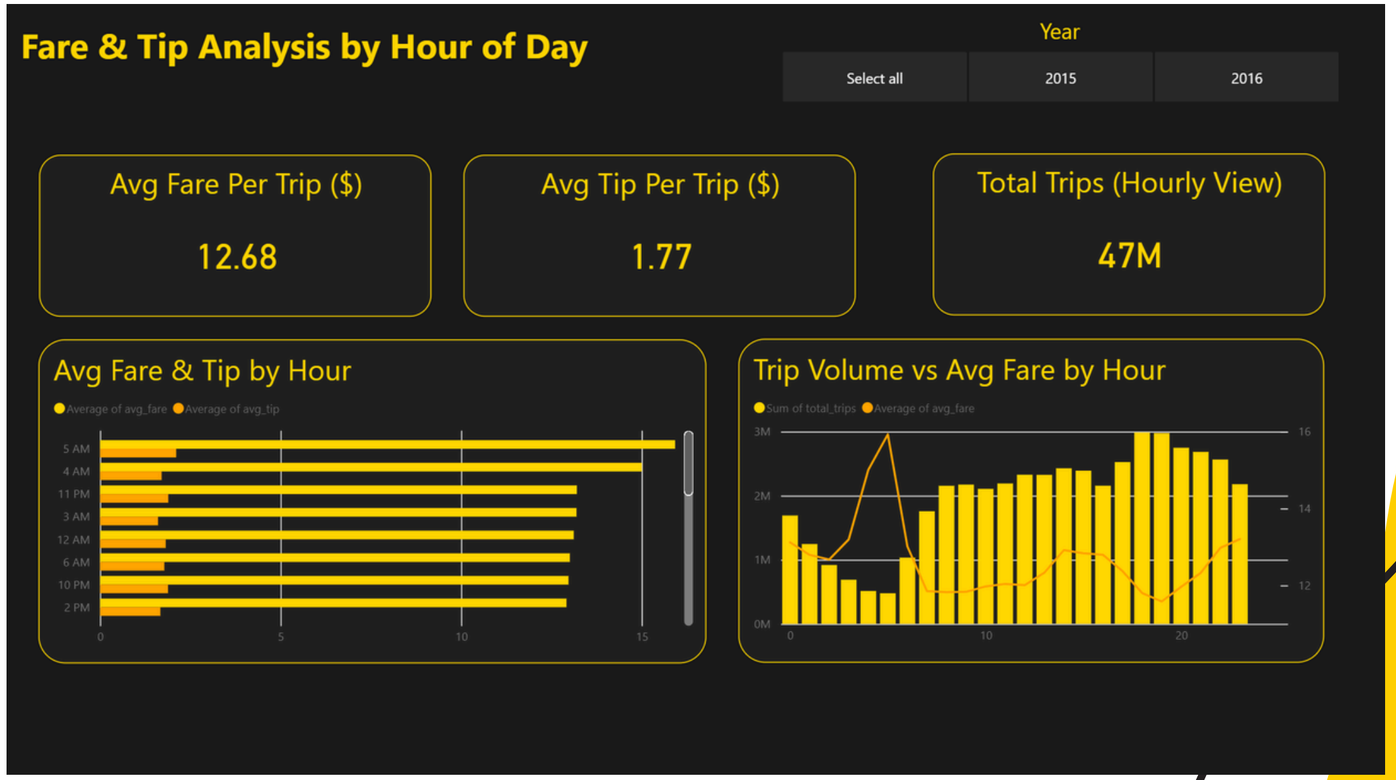
$12.68
Avg Fare Per Trip

$1.77
Avg Tip Per Trip

47M
Total Trips (Hourly View)

Key Observations

- Early morning (3–5 AM) has highest avg fare ~$15
- Evening hours show highest tip amounts
- Trip volume peaks at hours 18–20 (6–8 PM)
- Fare and volume show inverse relationship
- Busiest hours have lower avg fares (shorter trips)
- Avg tip is 14% of fare for credit card payments



DASHBOARD – PAYMENT TYPE BREAKDOWN

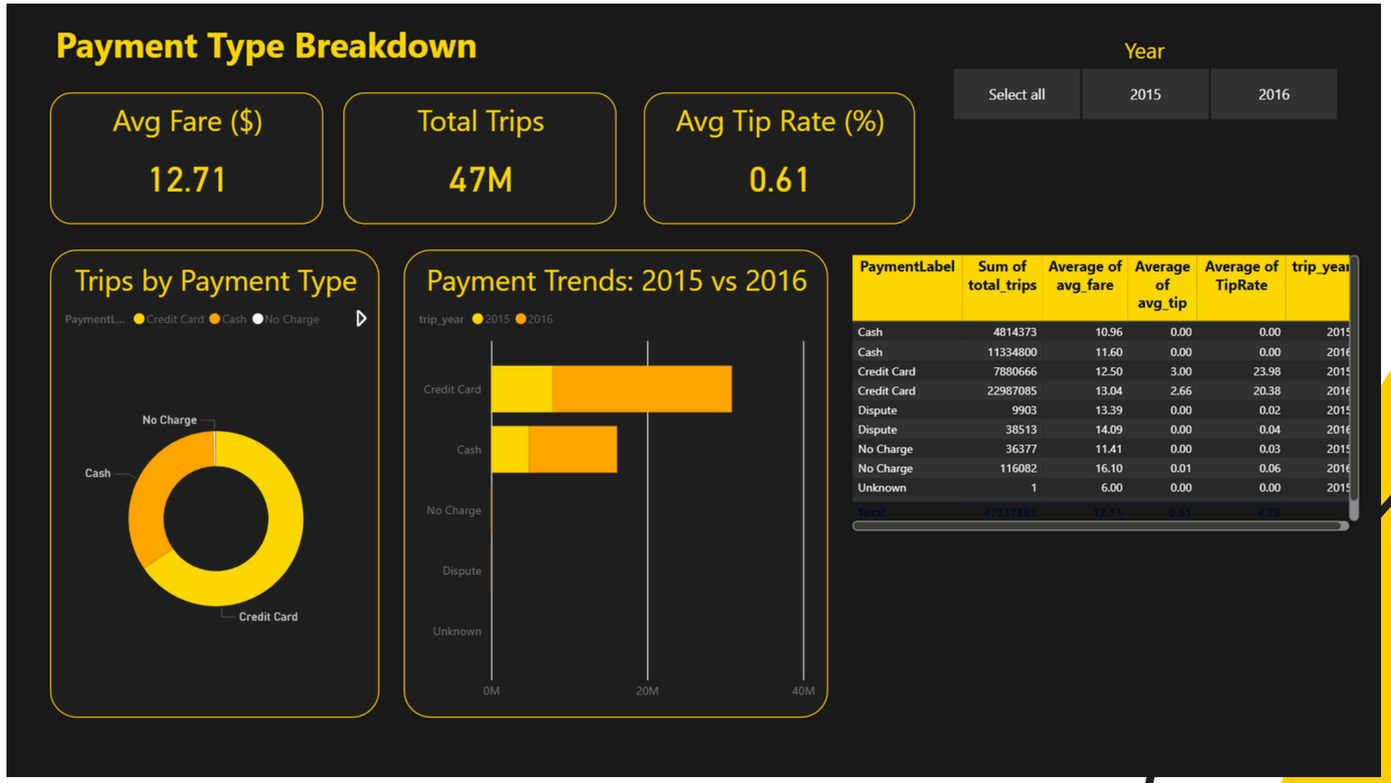
$12.71
Avg Fare

47M
Total Trips

0.61%
Avg Tip Rate

Key Observations

- Credit Card is the dominant payment method
- Credit card trips: 30.8M vs Cash: 16.1M
- Credit card avg fare: $12.50–$13.04
- Cash avg fare: $10.96–$11.60
- Cash trips show $0 tip (not recorded)
- 2016 shows growth in credit card usage vs 2015



PaymentLabel	Sum of total_trips	Average of avg_fare	Average of avg_tip	Average of TipRate	trip_year
Cash	4814373	10.96	0.00	0.00	2015
Cash	11334800	11.60	0.00	0.00	2016
Credit Card	7880666	12.50	3.00	23.98	2015
Credit Card	22987085	13.04	2.66	20.38	2016
Dispute	9903	13.39	0.00	0.02	2015
Dispute	38513	14.09	0.00	0.04	2016
No Charge	36377	11.41	0.00	0.03	2015
No Charge	116082	16.10	0.01	0.06	2016
Unknown	1	6.00	0.00	0.00	2015

DASHBOARD – NYC PICKUP LOCATION HEATMAP

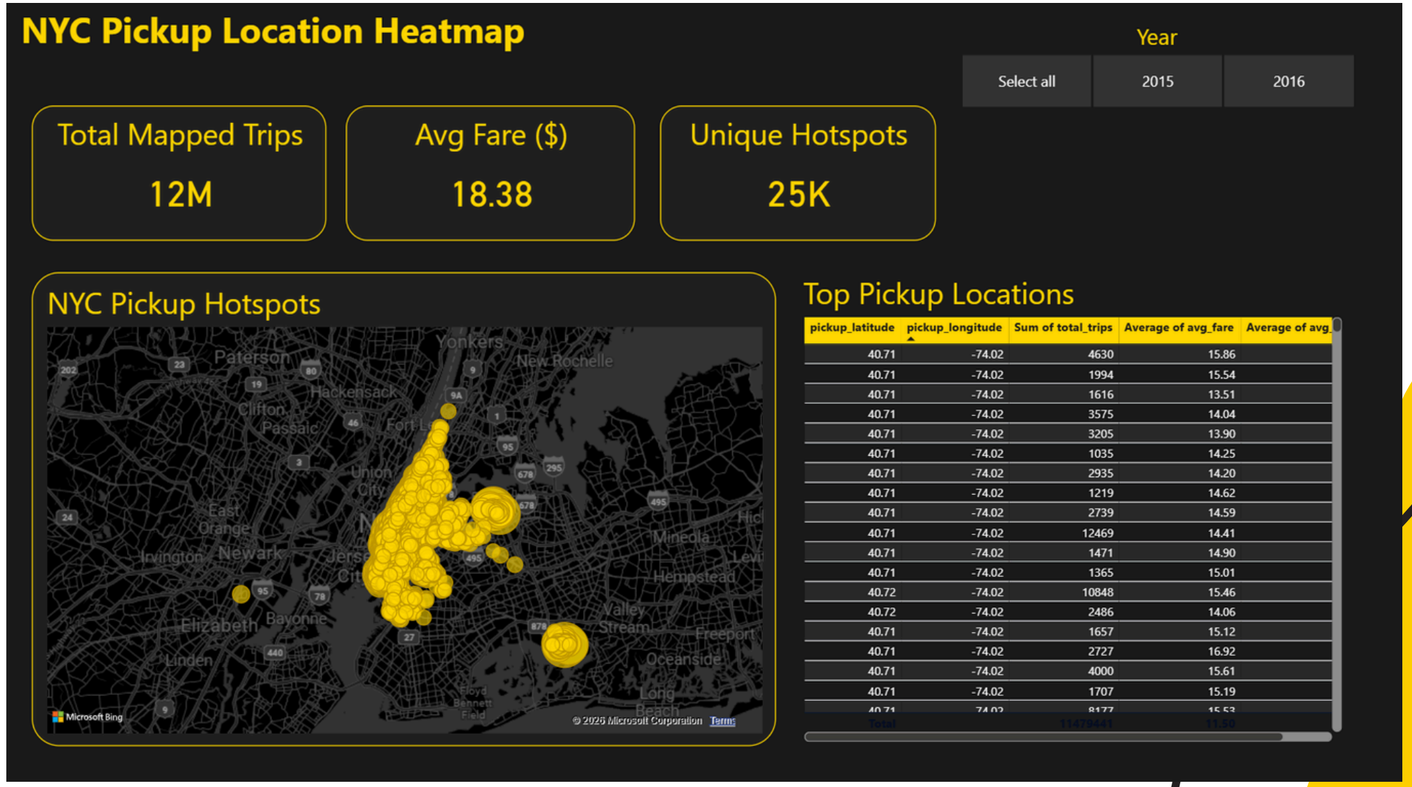
12M
Total Mapped Trips

$18.38
Avg Fare

25K
Unique Hotspots

Key Observations

- Pickups are heavily concentrated in Manhattan
- Top location: lat 40.71, lon -74.02 (4,630 trips)
- Airport routes show higher avg fares ($18+)
- Downtown Manhattan has densest bubble clusters
- 25,000 unique pickup coordinate combinations
- Outer boroughs show significantly fewer pickups



pickup_latitude	pickup_longitude	Sum of total_trips	Average of avg_fare	Average of avg
40.71	-74.02	4630	15.86	
40.71	-74.02	1994	15.54	
40.71	-74.02	1616	13.51	
40.71	-74.02	3575	14.04	
40.71	-74.02	3205	13.90	
40.71	-74.02	1035	14.25	
40.71	-74.02	2935	14.20	
40.71	-74.02	1219	14.62	
40.71	-74.02	2739	14.59	
40.71	-74.02	12469	14.41	
40.71	-74.02	1471	14.90	
40.71	-74.02	1365	15.01	
40.72	-74.02	10848	15.46	
40.72	-74.02	2486	14.06	
40.71	-74.02	1657	15.12	
40.71	-74.02	2727	16.92	
40.71	-74.02	4000	15.61	
40.71	-74.02	1707	15.19	

Key Insights & Findings

Peak Demand: Evening rush (6–8 PM) records 3M+ trips/hour — the busiest period for NYC yellow taxis

Fare Patterns: Early morning (3–5 AM) commands the highest avg fares (~$15–16) despite the lowest trip volume

Payment Shift: Credit card dominates at 30.8M trips vs 16.1M cash — digital payments overtook cash by 2016

Location: Manhattan accounts for the vast majority of pickups — outer boroughs see minimal yellow taxi activity

Tip Behaviour: Credit card tips avg $1.77/trip; cash trips record $0 (tips not captured in payment system)

Year-on-Year: 2016 shows increased trip volume vs 2015, with credit card usage growing significantly

Challenges & Solutions

Challenge: Authenticating Azure Synapse to securely access Blob Storage
Solution: Configured a SAS token credential and external data source so OPENROWSET could read CSV files directly from the raw-data container

Challenge: Raw table too large for Power BI (exceeded 1M row limit)
Solution: Created 5 aggregated Synapse SQL views — pre-summarized data (daily, hourly, by payment, by location) instead of raw rows

Challenge: Raw CSV data contained dirty records — negative fares, zero distances and invalid passengers
Solution: Applied TRY_CAST filtering in Synapse WHERE clause — removed 90K+ invalid rows, kept 12.7M clean records

Challenge: Azure IAM permissions — teammates couldn't access Synapse
Solution: Configured Storage Account Contributor role in Azure IAM for all team members

Conclusion & Future Work

What We Achieved

- End-to-end cloud data pipeline on Azure
- Processed 12.7M+ taxi trip records
- 6 optimized SQL views in Synapse
- 4-page interactive Power BI dashboard
- GitHub repo with full documentation

Future Work

- Add Apache Kafka for real-time trip streaming
- Expand dataset to 2017–2023 for trend analysis
- Build ML model to predict fare amounts
- Add weather data correlation analysis
- Deploy dashboard to Power BI Service (online)

Technologies Used

Azure Blob Storage || Azure Synapse Analytics || Serverless SQL || Power BI || DirectQuery || GitHub || Git LFS || NYC TLC Data

Thank You!

DAMG 7370 — NYC Yellow Taxi Data Pipeline | Spring 2025

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